



CSE6324-SOFTWARE VERIFICATION & VALIDATION

Topic: Hospital Management System

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Table of Contents

Table of Contents.....	2
1.0 Proposed Tool Identifications.....	3
2.0 Justification for the Project Environment.....	3
2.1 Type of Application.....	3
2.2 Ease of Setup and Execution.....	3
2.3 Support for Automation (Record & Playback).....	3
2.4 Programming Language Independence.....	4
3.0 Discussion of the 3 Evaluated Tools (Advantages and Pitfalls).....	4
4.0 Workflow Evidence and Execution.....	5
4.1 TestCase Studio.....	5
4.2 Katalon Recorder.....	6
4.3 Google Lighthouse.....	8

1.0 Proposed Tool Identifications

For the comprehensive testing of the Hospital Management System (HMS), our group proposes a multi-tool "Testing Suite" approach utilizing three distinct tools to cover different aspects of the testing lifecycle: **TestCase Studio**, **Katalon Recorder**, and **Google Lighthouse**.

Rather than relying on a single tool, this suite allows us to efficiently document manual test steps and locators (TestCase Studio), perform lightweight automated UI execution (Katalon Recorder), and conduct automated non-functional performance auditing (Google Lighthouse).

2.0 Justification for the Project Environment

2.1 Type of Application

The HMS is a monolithic web-based application utilizing HTML, CSS, JavaScript, and PHP. All three proposed tools interact directly with the Document Object Model (DOM) through the browser, making them perfect for black-box frontend testing without needing to alter the application's source code.

2.2 Ease of Setup and Execution

Setting up robust automation environments (like Selenium WebDriver with Java or PHPUnit) requires configuring heavy IDEs and downloading drivers. In contrast, TestCase Studio and Katalon Recorder are simply added as browser extensions, and Google Lighthouse is natively built into Chrome. Setup takes seconds, and tests can execute instantly on our local XAMPP environment.

2.3 Support for Automation (Record & Playback)

This suite provides full-spectrum automation support. TestCase Studio automates the writing of test cases by capturing plain-English steps and XPath locators. Katalon Recorder features a highly accurate "Record and Playback" utility for functional UI testing. Finally, Google Lighthouse automates the auditing process for page load speeds and accessibility.

2.4 Programming Language Independence

Because these tools record frontend interactions and scan the rendered DOM, they are completely independent of our PHP backend. This allows all group members to generate test documents, execute UI scripts, and run performance audits regardless of their programming expertise.

3.0 Discussion of the 3 Evaluated Tools (Advantages and Pitfalls)

To ensure we selected the best tool, we compared Selenium IDE against two other easy-to-use alternatives: Katalon Studio (Automated UI) and TestRail (Test Management).

To ensure comprehensive testing coverage across both functional and non-functional requirements, our group evaluated and selected the following three tools:

Tool 1: TestCase Studio (Test Documentation & Locators)

- **Advantages:** Instantly translates manual clicks into plain-English test steps and automatically generates highly accurate XPath and CSS selectors for every element. It easily exports to Excel, perfectly bridging the gap between manual testing and automated script preparation.
- **Pitfalls:** It is purely a recording and documentation tool; it does not have a playback engine to automatically execute tests or verify pass/fail statuses.

Tool 2: Katalon Recorder (Lightweight UI Execution)

- **Advantages:** A lightweight browser extension that provides instant "Record and Playback" capabilities. It automatically captures UI interactions and allows testers to play them back at high speeds, providing visual pass (green) and fail (red) execution logs.
- **Pitfalls:** It lacks advanced reporting dashboards (no graphical PDF reports natively) and can be difficult to scale for massive, complex data-driven testing compared to heavy desktop applications.

Tool 3: Google Lighthouse (Automated Performance Auditing)

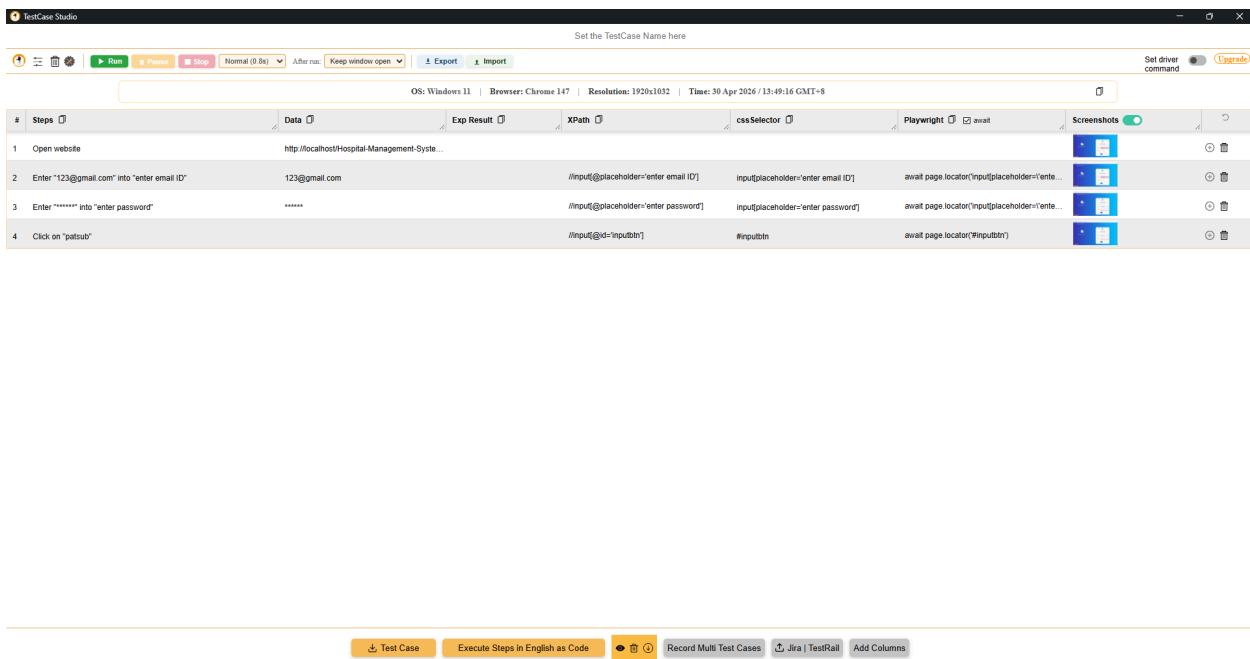
- **Advantages:** Requires zero installation as it is built directly into Chrome Developer Tools. With a single click, it runs automated audits against the web page to test performance load times and accessibility, generating a highly visual, professional report with scoring metrics.
- **Pitfalls:** It is strictly a non-functional testing tool. It measures *how well* the site runs (speed, SEO, accessibility) but cannot test business logic, user authentication flows, or database interactions.

4.0 Workflow Evidence and Execution

4.1 TestCase Studio

Step 1: Test Case Generation and Element Locator Capture

The workflow begins using the TestCase Studio web recorder. As the tester navigates the HMS environment (e.g., inputting an email, password, and clicking "Login"), the tool captures these UI interactions and generates the corresponding automation locators (XPath and CSS Selectors).



The TestCase Studio interface successfully captures plain-English test steps alongside the required XPath and CSS selectors during a patient login attempt.

Step 2: Test Case Export and Documentation

Once the session is complete, the recorded data is exported. By clicking the "Test Case" download function, a structured spreadsheet is generated containing the steps and expected data, serving as official test documentation.

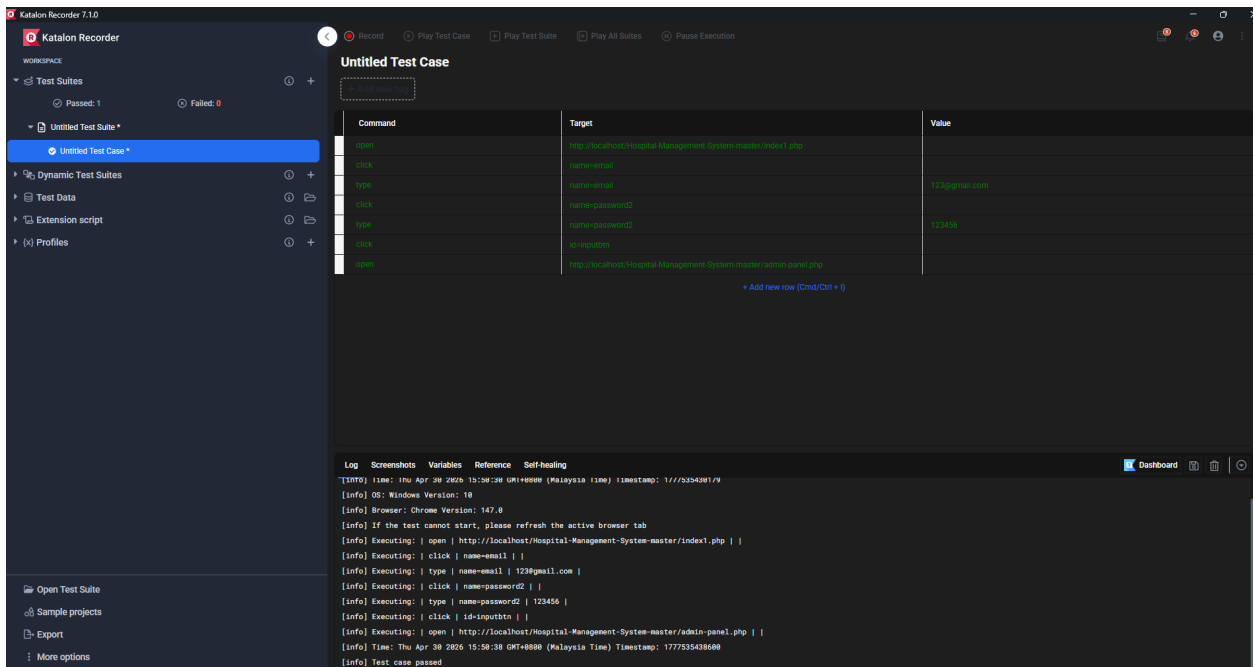
A	B	C	D	E	F	G	H	I	J	K
	OS: Windows 11 Browser: Chrome 147 Resolution: 1536x816 Time: 30 Apr 2026 / 13:58:27 GMT+8									
#	Step	Data	Exp Result	XPath	cssSelector					
1	Open webs	http://localhost/Hospital-Management-System-master/index1.php								
2	Enter "123	123@gmail.com		//input[@p	input[placeholder='enter email ID']					
3	Enter "****	*****		//input[@p	input[placeholder='enter password']					
4	Click on "pat	sub		//input[@ic	#inputbtn					

The exported Excel spreadsheet generated by TestCase Studio, perfectly organizing the test steps and locators for project documentation.

4.2 Katalon Recorder

Step 1: Test Execution & Verification

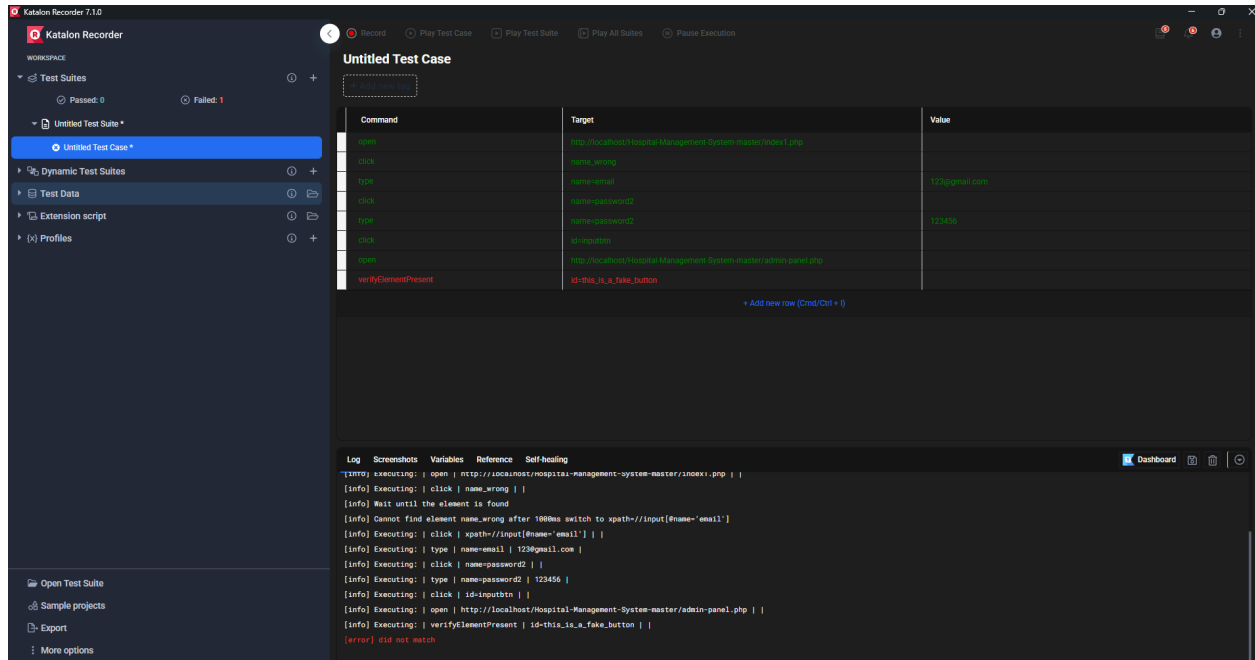
The tester records a workflow (e.g., navigating to the index, entering credentials, and submitting) in Katalon Recorder. Upon commanding the tool to "Play", it automatically executes the browser actions. Successfully executed steps are highlighted in green, verifying the system's stability.



Katalon Recorder successfully executes the login script, highlighting all passed steps in green and displaying a success message in the log viewer.

Step 2: Defect Tracking and Reporting

To demonstrate defect tracking, an incorrect target ID (e.g., an invalid button) is intentionally targeted via a `verifyElementPresent` command. Katalon Recorder halts execution, highlights the broken step in red, and logs the specific error trace.

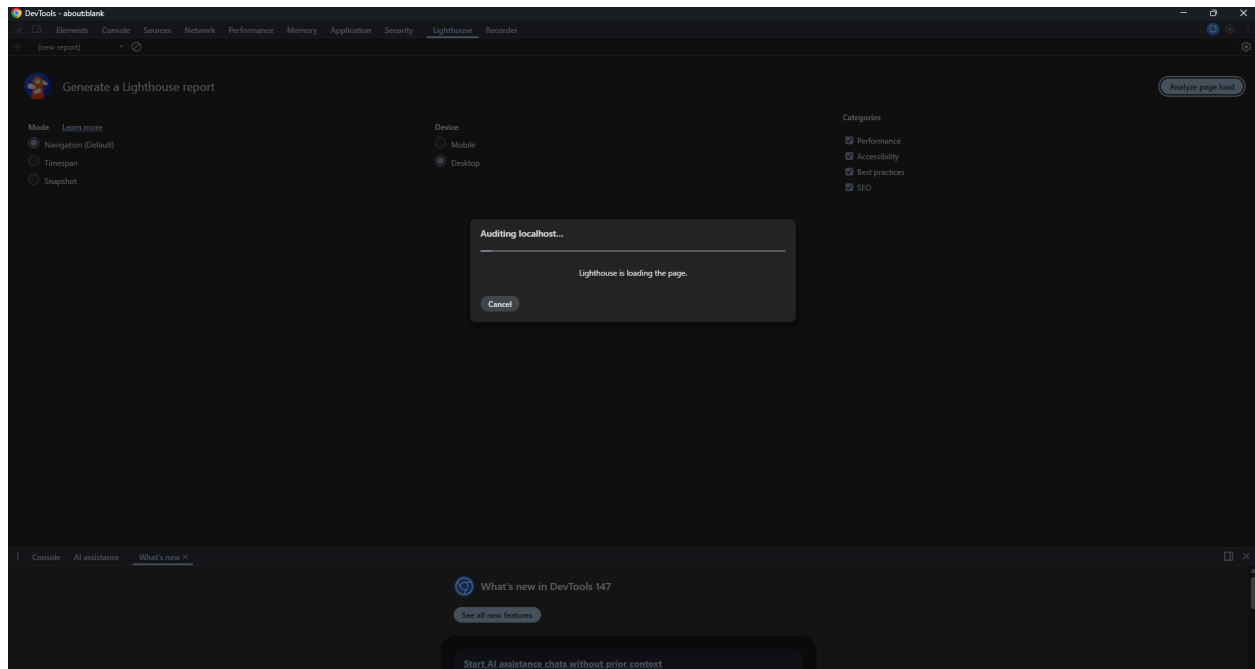


Katalon Recorder catching a defect. The tool highlights the failed step in red and prints a clear "[error] did not match" message in the console.

4.3 Google Lighthouse

Step 1: Automated Audit Execution

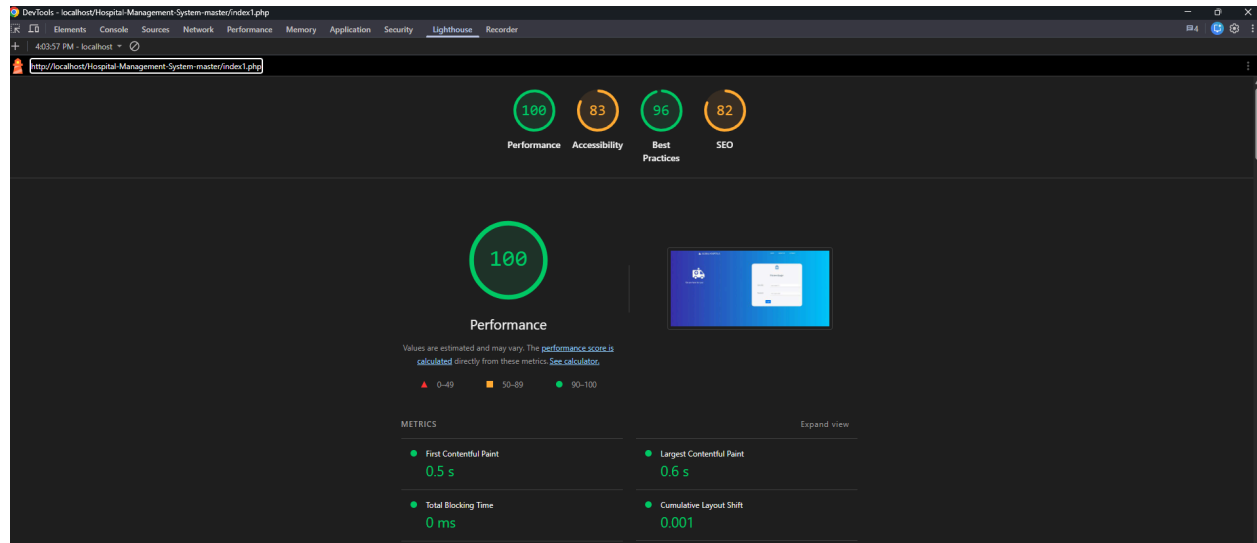
The tester navigates to the HMS portal and initializes the Lighthouse tool via the Chrome Developer Tools environment. The tool is commanded to run an automated analysis on the page's load performance, simulating a page load and gathering metrics.



Initializing the automated performance audit on the local HMS environment using Chrome's built-in Lighthouse DevTools feature.

Step 2: Report Generation and Metrics

Upon completion, Lighthouse automatically generates a comprehensive diagnostic report. The dashboard scores the application across categories such as Performance, Accessibility, and Best Practices (out of 100), providing developers with immediate non-functional feedback.



The final generated Lighthouse report showed an excellent Performance score of 100, detailing rendering times and layout shifts for the HMS index page.